Uniform plane wave propagation

Dr. Sampath Kumar
Associate Professor
Department of ECE
MIT, Manipal

REVISION: Wave equation

$$\nabla^2 \vec{E} = -\omega^2 \mu \left(\epsilon - j\frac{\sigma}{\omega}\right) \vec{E}$$

$$\nabla^2 \vec{E} = -\omega^2 \mu (\epsilon' - j\epsilon'') \vec{E}$$

$$\epsilon' = \epsilon_0 \epsilon_r$$

$$\epsilon'' = \frac{\sigma}{\omega}$$

In compact form $\nabla^2 \vec{E} = -k^2 \vec{E}$

where $jk = \sqrt{-\omega^2 \mu (\epsilon' - j\epsilon'')}$

Using the calculus one can write it as $\boxed{jk = \alpha + j\beta}$

where

$$\alpha = \sqrt{\frac{\omega^2 \mu \epsilon'}{2}\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}} - 1\right]}$$

and

$$\beta = \sqrt{\frac{\omega^2 \mu \epsilon'}{2}\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{1/2} + 1\right]}$$

$$\eta = \frac{E_x}{H_y} = \frac{\omega\mu}{k} = \frac{j\omega\mu}{\alpha + j\beta}$$

Wave propagation in partially conducting medium (Lossy dielectric medium)

Here $\sigma \neq 0 \quad \therefore \epsilon'' = \frac{\sigma}{\omega} \neq 0$

For direction of propagation in +ve z direction

$$\vec{E} = \hat{a}_x E_0 e^{j(\omega t - kz)}$$

$$\vec{E} = \hat{a}_x E_0 e^{-jkz} \cdot e^{j\omega t}$$

$$= \hat{a}_x E_0 e^{-\alpha z} \cdot e^{-j\beta z} \cdot e^{j\omega t}$$

$$jk = \alpha + j\beta$$

$$-jk = -\alpha - j\beta$$

$$\vec{E} = \hat{a}_x E_0 e^{-\alpha z} e^{j(\omega t - \beta z)}$$

Field attenuates exponentially

wave in +ve z direction

$\Downarrow$

conductive loss

α = Attenuation coefficient (nepers/m)

β = phase constant (rad/m)

k = propagation constant

K, α, β

Wave impedance η :-

$$\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{-j\sqrt{-\omega^2\mu(\epsilon' - j\epsilon'')}}$$

→ Multiply both numerator & denominator by $\sqrt{\mu}$

→ Take j inside square root

→ Substitute for ϵ' & ϵ''

We get $$\eta = \frac{\omega\mu\sqrt{\mu}}{\sqrt{\omega^2\mu^2\left(\epsilon - \frac{\sigma}{\omega}\right)}}$$

$$\eta = \sqrt{\frac{\mu/\epsilon}{\left(1 - j\frac{\sigma}{\omega\epsilon}\right)}}$$

Converting into polar form

$$= \sqrt{\frac{\mu/\epsilon}{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2\right]^{1/2}}} \cdot \exp\left[\underbrace{\frac{j}{2}\tan^{-1}\left(\frac{\sigma}{\omega\epsilon}\right)}_{\text{phase}}\right]$$

Consider expression for $\vec{H}$

$$\vec{E} = \hat{a}_x E_0 e^{-\alpha z} e^{j(\omega t - \beta z)}$$

$$\vec{H} = \hat{a}_y \frac{E_0}{\sqrt{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2\right]^{1/2}}} \cdot e^{-\alpha z} e^{j\left(\omega t - \beta z - \frac{1}{2}\tan^{-1}\left(\frac{\sigma}{\omega\epsilon}\right)\right)}$$

$$\vec{H} = \hat{a}_y \frac{E_0}{\sqrt{\left[1+\left(\frac{\sigma}{\omega\epsilon}\right)^2\right]^{1/2}}} \cdot e^{-\alpha z} e^{j\left(\omega t - \beta z - \frac{1}{2}\tan^{-1}\left(\frac{\sigma}{\omega\epsilon}\right)\right)}$$

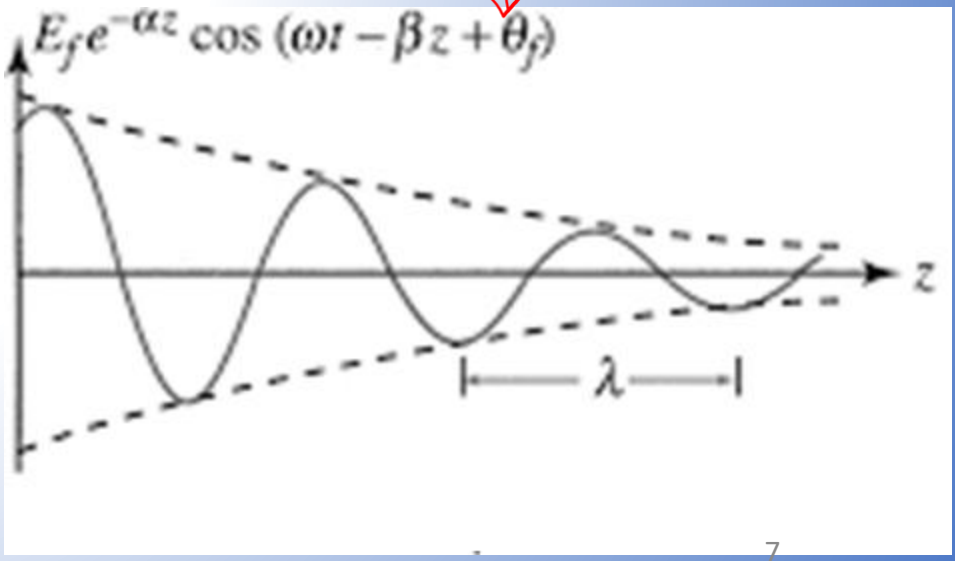
But WkT

$$E = \hat{a}_x E_0 e^{-\alpha z} e^{j(\omega t - \beta z)}$$

Compare these two equations

⇒ 1) Both waves attenuates as moving in z direction

2) Both are out of phase



Proof V = fλ

Further, $V = \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\frac{\omega^2 \mu \epsilon'}{2}\left[1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2\right]^{1/2} + 1}}$

$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{f\sqrt{\frac{\mu \epsilon'}{2}\left[1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2\right]^{1/2} + 1}}$

$\Rightarrow V = f\lambda$

Wave propagation in perfect dielectric medium

$\sigma = 0 \Rightarrow \epsilon'' = 0$

$\vec{E} = \hat{a}_x E_0 e^{j(\omega t - \beta z)}$

$\eta = \frac{\omega\mu}{k} = \frac{j\omega\mu}{\alpha + j\beta} = \frac{\omega\mu}{\beta} \quad (\because \alpha = 0)$

But $\beta = \sqrt{\frac{\omega^2\mu\epsilon'}{2}\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{1/2}+1\right]}$

Put $\epsilon'' = 0$ we get

$\beta = \omega\sqrt{\mu\epsilon}$

$\eta = \frac{\omega\mu}{\beta} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}}$ [For free space 377Ω]

$\vec{H} = \hat{a}_y \frac{E_0}{\sqrt{\mu/\epsilon}} e^{j(\omega t - \beta z)}$

$v = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu\epsilon}}$ [For free space $v = c$]

$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{\mu\epsilon}} = \frac{1}{f\sqrt{\mu\epsilon}}$

E and H are in phase

Wave propagation in good conductor

$\sigma \gg \omega\epsilon \Rightarrow \frac{\sigma}{\omega} = \epsilon'' \gg \epsilon = \epsilon'$

$\Rightarrow$ Imaginary part of complex permitivity becomes significant

$$\alpha = \sqrt{\frac{\omega^2\mu\epsilon'}{2}\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}}-1\right]}$$

$\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}}-1\right] \approx \left(\frac{\epsilon''}{\epsilon_1}\right)$

$$\therefore \alpha = \sqrt{\frac{\omega^2\mu\epsilon'}{2}\cdot\frac{\epsilon''}{\epsilon'}} = \omega\sqrt{\frac{\mu}{2}\cdot\frac{\sigma}{\omega}}$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}} \Rightarrow \alpha = \sqrt{\pi f\mu\sigma}$$

Define $\frac{1}{\sqrt{\pi f\mu\sigma}} = \delta$ = skin depth = Depth of penetration

$\therefore \alpha = \frac{1}{\delta}$

$$\beta = \sqrt{\frac{\omega^2\mu\epsilon'}{2}\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}}+1\right]}$$

Since ϵ'' is very large we can write

$$\beta = \alpha = \sqrt{\frac{\omega\mu\sigma}{2}} = \frac{1}{\delta}$$

Now $\eta = \frac{\omega\mu}{k} = \frac{j\omega\mu}{\alpha + j\beta} = \frac{j\omega\mu}{\sqrt{\frac{\omega\mu\sigma}{2}}(1+j)}$

η

Using polar form

$$\eta = \frac{j\sqrt{\omega\mu}}{\sqrt{\frac{\sigma}{2}}\sqrt{1+1}\cdot e^{j\frac{\pi}{4}}} = \frac{j\sqrt{\omega\mu}}{\sqrt{\sigma}}\cdot e^{-j\frac{\pi}{4}} = \frac{\sqrt{\omega\mu}}{\sqrt{\sigma}}\cdot e^{-j\frac{\pi}{4}}\cdot e^{j\frac{\pi}{2}}$$

$\eta = \sqrt{\frac{\omega\mu}{\sigma}}\cdot e^{j\frac{\pi}{4}} \Rightarrow$ Wave impedance is complex & phase is $\pi/4$

$$\vec{E} = \hat{a}_x E_0 e^{-\alpha z} \cdot e^{j(\omega t - \beta z)}$$

Skin depth

$$\vec{H} = \hat{a}_y \frac{E_0}{\sqrt{\omega\mu/\sigma}} e^{j(\omega t - \beta z - \pi/4)}$$

$\vec{E}$ & $\vec{H}$ will have phase diff. of $\frac{\pi}{4} = 45^\circ$

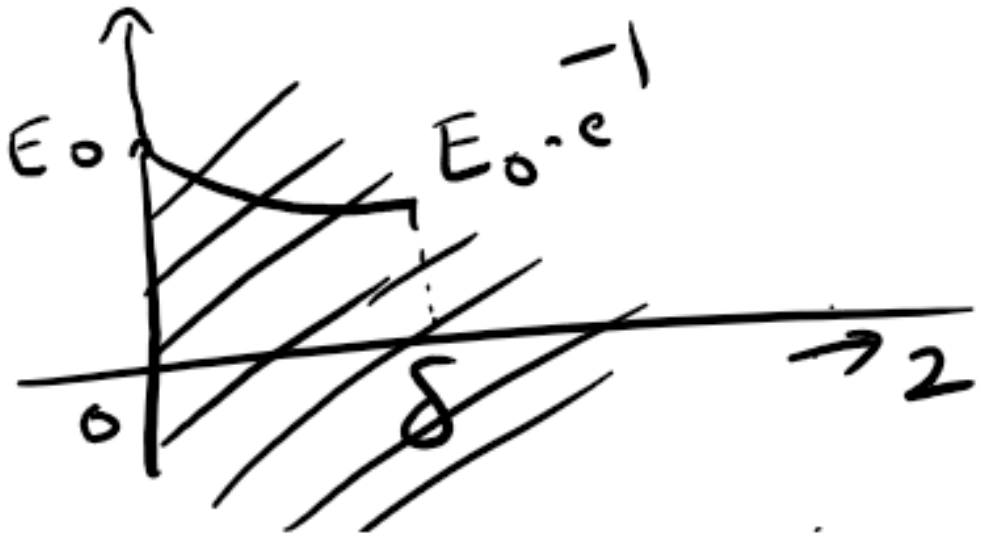
(A special case of partially conducting medium)

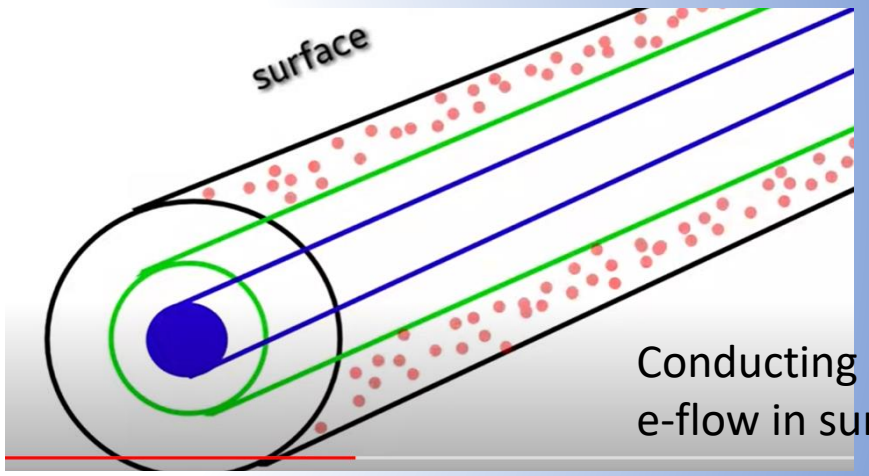
$$V = \frac{\omega}{\beta} = \omega\delta \; ; \; \lambda = 2\pi\delta$$

Meaning of Skin depth

$$\vec{E} = \hat{a}_x E_0 e^{-\alpha z} \cdot e^{j(\omega t - \beta z)}$$

$$\vec{E} = \hat{a}_x E_0 e^{-\frac{z}{\delta}} \cdot e^{j(\omega t - \beta z)}$$





Conducting in layers
e-flow in surfaces

Large δ → lower attenuation
small δ → higher attenuation


Thank
you